



YISHUN INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE
NAME

CG

INDEX NO

BIOLOGY

9744/01

Paper 1 Multiple Choice

16 Sep 2022

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not use staples, paper clips, glue or correction fluid/tape.
Write your name, index no. and CG on this cover page and OTAS provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

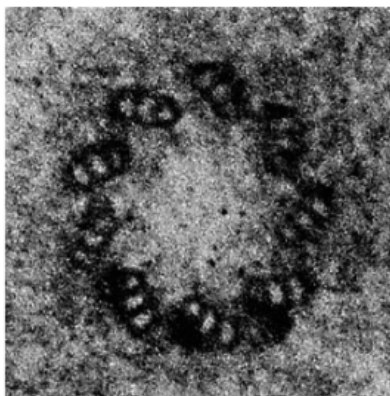
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

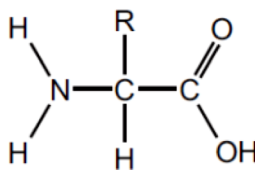
This document consists of **21** printed pages and **1** blank page.

- 1 The electron micrograph shows a structure found in the cytoplasm of an animal cell.



What is this cell structure?

- A centriole
 - B lysosome
 - C ribosome
 - D vesicle
- 2 The diagram represents an amino acid.



R represents a variable side chain.

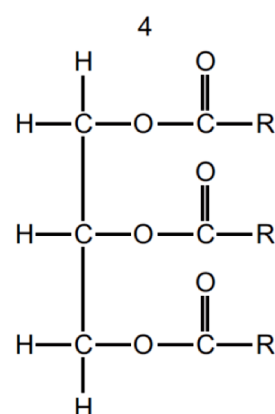
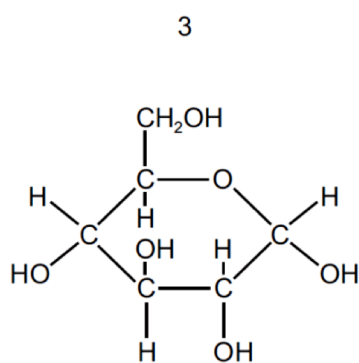
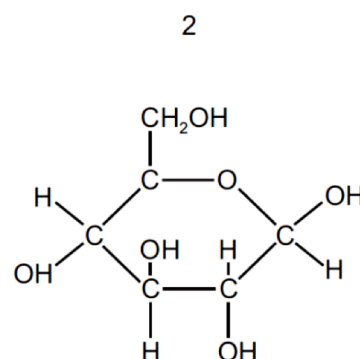
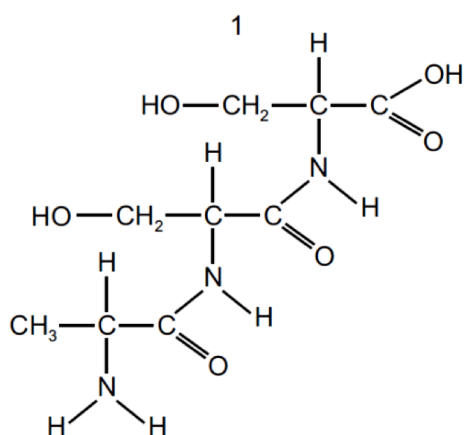
Which of the following is **not** a possible side chain?

- A CH_3
- B $\text{CH}_2\text{CH}_2\text{SCH}_3$
- C CH_2CONH_2
- D $\text{HOCH}_2\text{CH}(\text{OH})\text{CH}_2\text{OH}$

- 3 A student carried out four tests for biological molecules on a solution. The results are shown in the table.

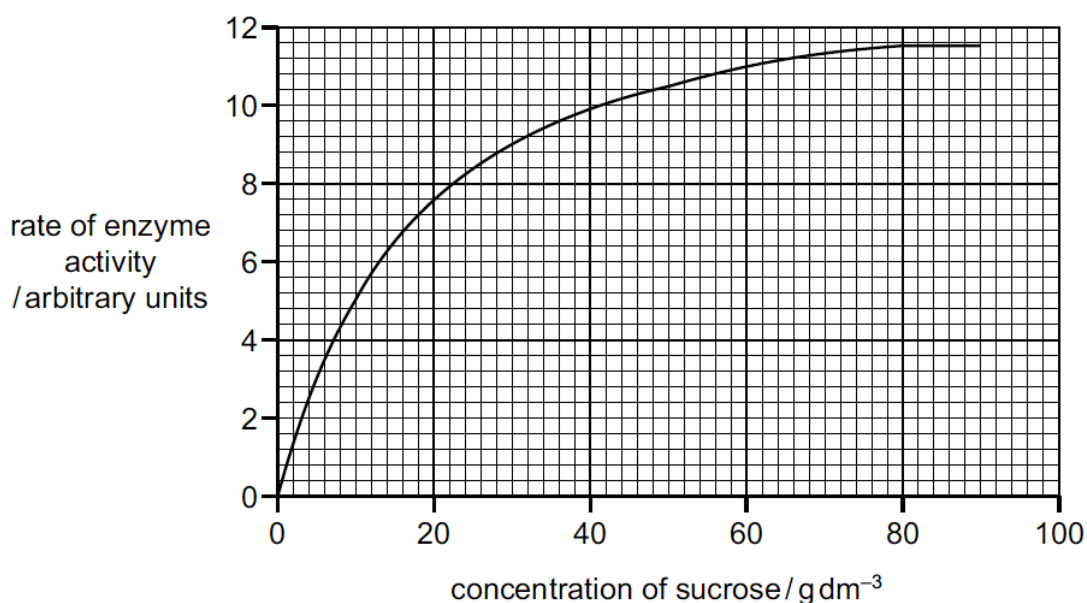
test for biological molecule	observation
iodine solution	orange
biuret	blue
Benedict's	orange
emulsion	clear

Which molecules are present in this solution?



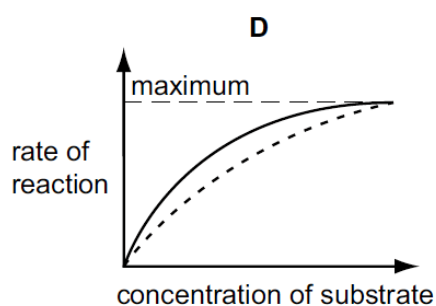
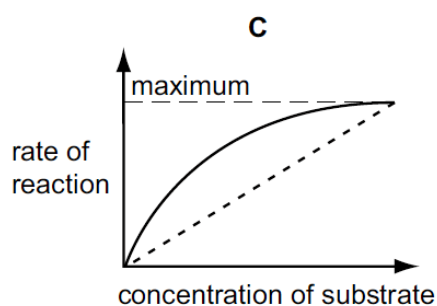
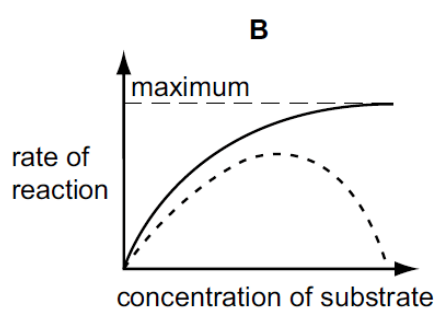
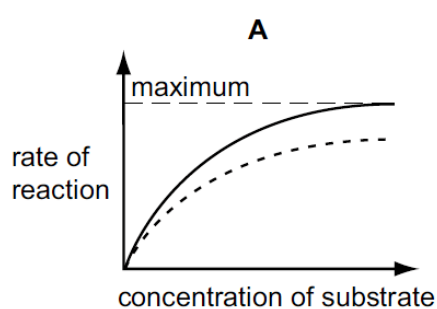
- A** 1 and 3
B 1 and 4
C 2 and 3
D 3 and 4

- 4 The graph shows the rate of activity of the enzyme sucrase plotted against the concentration of sucrose.



Why does the rate of enzyme activity remain constant from 80 – 90 g dm^{-3} .

- A All the enzyme has been inhibited.
 B All the substrate has been used up.
 C The concentration of the enzyme is limiting the rate.
 D The concentration of the substrate is limiting the rate.
- 5 Which graph represents the action of a non-competitive inhibitor?



key

— without inhibitor

- - - with inhibitor

6 The following are all observations about cell surface membranes.

- 1 Phospholipids labelled with radioactive phosphate groups change position over time.
- 2 Proteins labelled with fluorescent dyes change position over time.
- 3 Lectins are proteins that bind to polysaccharides. Lectins only attach to the outside of cell surface membranes.
- 4 Electron microscope images of membranes fractured by freezing shows a large number of regularly arranged particles interrupted by larger particles.
- 5 Some proteins can easily be removed from the membrane by changing the ionic balance.
- 6 Some proteins can only be separated from the membrane by disrupting the membrane with detergent.

Which observations provide evidence for the fluid mosaic model of cell surface membranes?

- A** 1, 2 and 4
- B** 1, 2 and 6
- C** 3, 4 and 5
- D** 3, 5 and 6

7 Some stem cells divide and give rise to phagocytes.

Where in the human body do these stem cells divide?

- 1 blood
 - 2 bone marrow
 - 3 lymph nodes
- A** 1, 2 and 3
 - B** 1 and 3 only
 - C** 2 only
 - D** 3 only

8 Which statement(s) about RNA is / are correct?

- 1 It is less stable than DNA as it contains a ribose sugar that lacks a 2' OH group.
- 2 It is a single stranded polymer of purine and pyrimidine joined by phosphodiester bonds.
- 3 It is synthesised in the 5' to 3' direction where the 5'-phosphate group of the growing RNA strand is joined to the 3'-hydroxyl group of an incoming nucleotide.
- 4 The function of ribosomal RNA is to catalyse the formation of peptide bonds.

- A** 4 only
- B** 1 and 3 only
- C** 2 and 3 only
- D** 1, 3 and 4 only

9 The following statements list the processes that occur during translation.

- 1 The large subunit of the ribosome binds and forms the translation initiation complex.
- 2 The second amino acyl-tRNA complex now binds to mRNA at the "A" site of the ribosome.
- 3 The small ribosomal subunit, with initiator tRNA bound, binds to the 5' cap of the mRNA and scans for the first start codon.
- 4 Soluble protein called release factor recognises the stop codon and binds at the "A" site.
- 5 Formation of a peptide bond between the first and the second amino acids by peptidyl transferase.
- 6 The second amino acyl-tRNA complex moves from the "A" site to the "P" site.

Using the information provided above, deduce the order in which these processes occur.

- A** 1 → 3 → 2 → 5 → 6 → 4
- B** 1 → 3 → 2 → 6 → 5 → 4
- C** 3 → 1 → 2 → 5 → 6 → 4
- D** 3 → 1 → 2 → 6 → 5 → 4

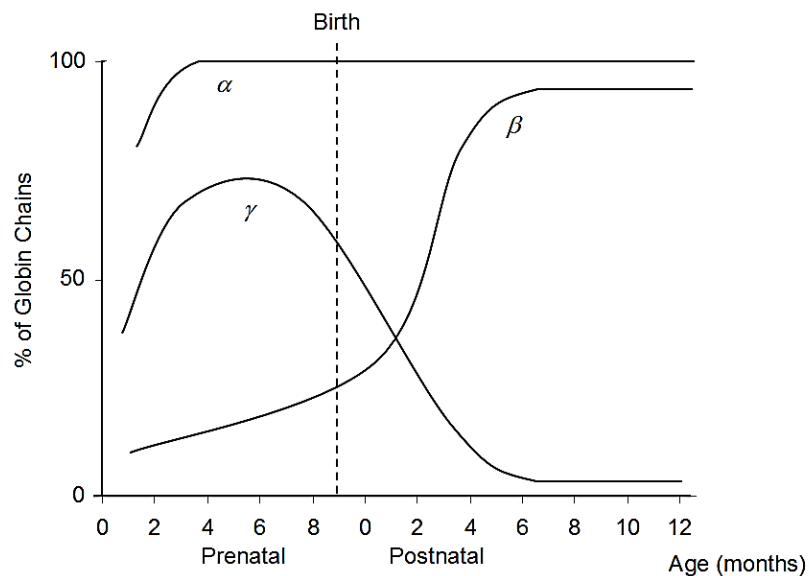
- 10 The non-template strand of part of a DNA molecule has the sequence 5' GAATTA 3'.

Which row is correct for this part of the corresponding template and tRNA strands?

	the sequence of the template strand	the anticodons of tRNA used in translation
A	5' CTTAAT 3'	5' CUU 3' and 5' AAU 3'
B	5' CTTAAT 3'	5' UAA 3' and 5' UUC 3'
C	5' TAATTC 3'	5' CUU 3' and 5' AAU 3'
D	5' TAATTC 3'	5' UAA 3' and 5' UUC 3'

- 11 The globin gene family in humans consists of α , β and γ genes. These genes code for the globin chains that make up haemoglobin and are expressed at different levels during different developmental stages.

The graph shows the expression of the various globin chains during the prenatal (fetal) and postnatal (after birth) periods.



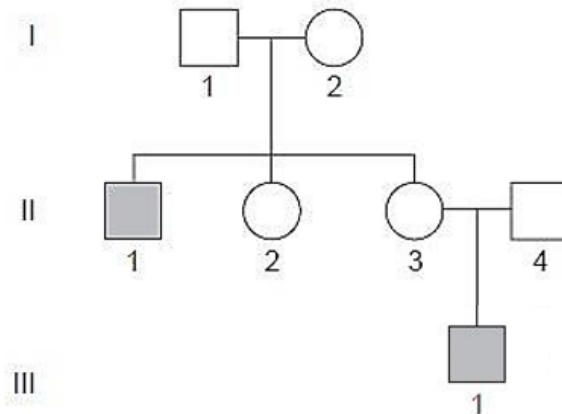
Which statement **cannot** account for the differences in the levels of expression of globin chains?

- A A growth factor triggers the expression of a transcription factor that increases the rate of β -globin gene expression during the postnatal period.
- B Alternative splicing occurs in the mRNA of the α -globin and β -globin genes, resulting in differences in the rate of expression of globin chains during the prenatal period.
- C Methyl groups are added to regulatory sequences of γ -globin genes during the prenatal period, allowing for some proteins to bind.
- D The shortening of poly(A) tail in the mRNA of γ -globin genes reduces its stability, resulting in a decrease in the rate of expression of γ -globin chains during the prenatal period.

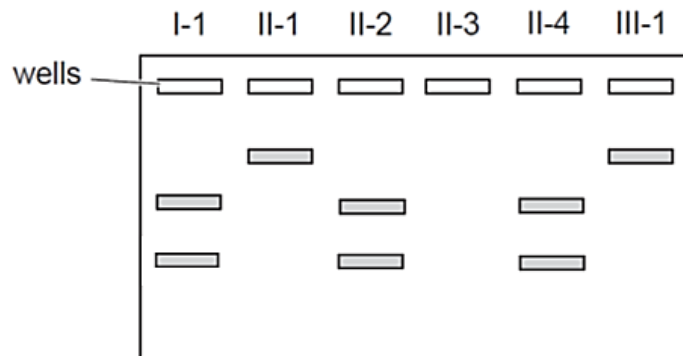
- 12** Menkes' Disease in humans is characterised by sparse and wiry hair, growth failure and deterioration of the nervous system. Onset of the Menkes' syndrome usually occurs during infancy.

Restriction digestion was carried out on DNA samples taken from a family in which this X-linked disorder was present. The DNA fragments obtained were subjected to gel electrophoresis.

The family pedigree is shown below.



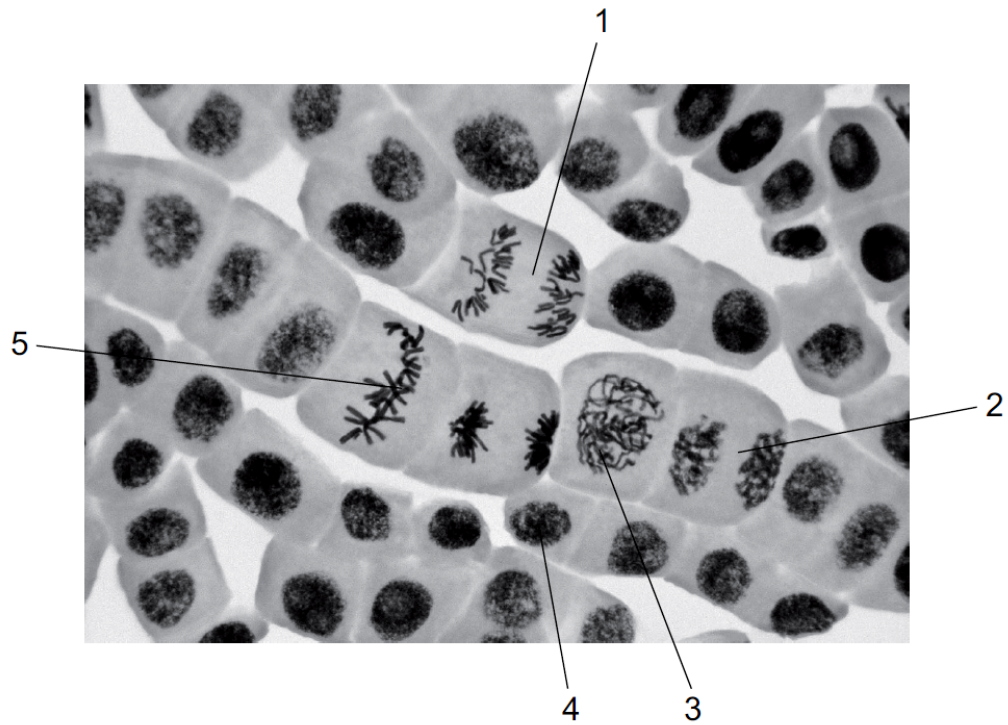
The figure below shows the results obtained after gel electrophoresis.



What would be the band pattern of individual II-3?

A **B** **C** **D**

13 The photomicrograph shows cells in different stages of mitosis.

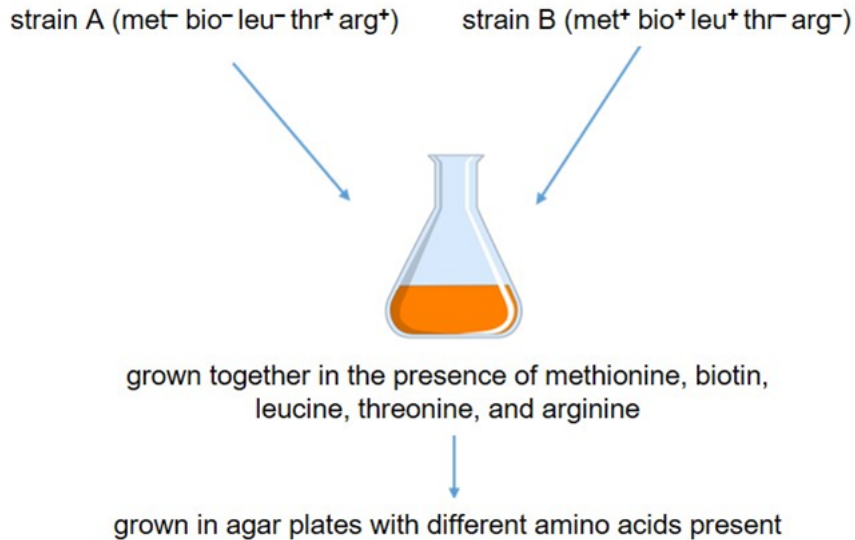


In which order do these stages occur?

- A** 3 → 1 → 5 → 2 → 4
- B** 3 → 5 → 2 → 1 → 4
- C** 4 → 3 → 5 → 1 → 2
- D** 4 → 5 → 1 → 2 → 3

- 14 Essential amino acids e.g. methionine (met), biotin (bio), leucine (leu), threonine (thr) and arginine (arg) are critical for the bacteria cells to survive and replicate. Some bacteria carried the genes required for the synthesis of the amino acid (indicated by "+") while others with the genes (indicated by "-") will take up these amino acids from the culture medium.

An experiment involving two strains of bacteria were conducted to investigate gene transfer.



The results of the investigation are summarised in the table below.

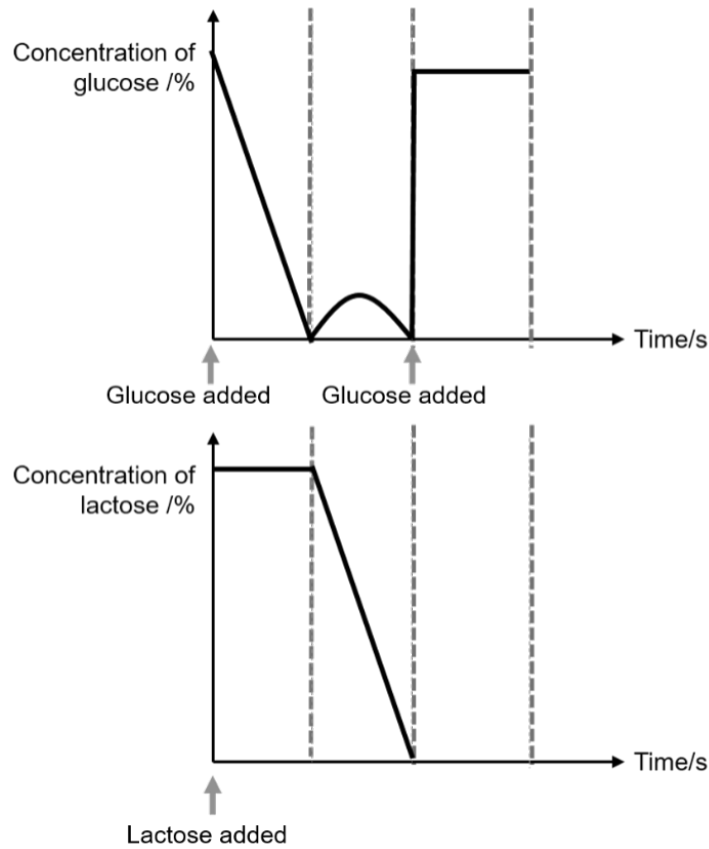
flask	bacterial strain grown	amino acid present in agar plate					presence of bacteria colonies
		met	bio	leu	thr	arg	
1	A	✓	✗	✓	✗	✗	no
2	B	✗	✗	✗	✓	✓	yes
3	A + B	✗	✗	✗	✗	✗	yes

Which of the following process(es) could explain the results in flask 3?

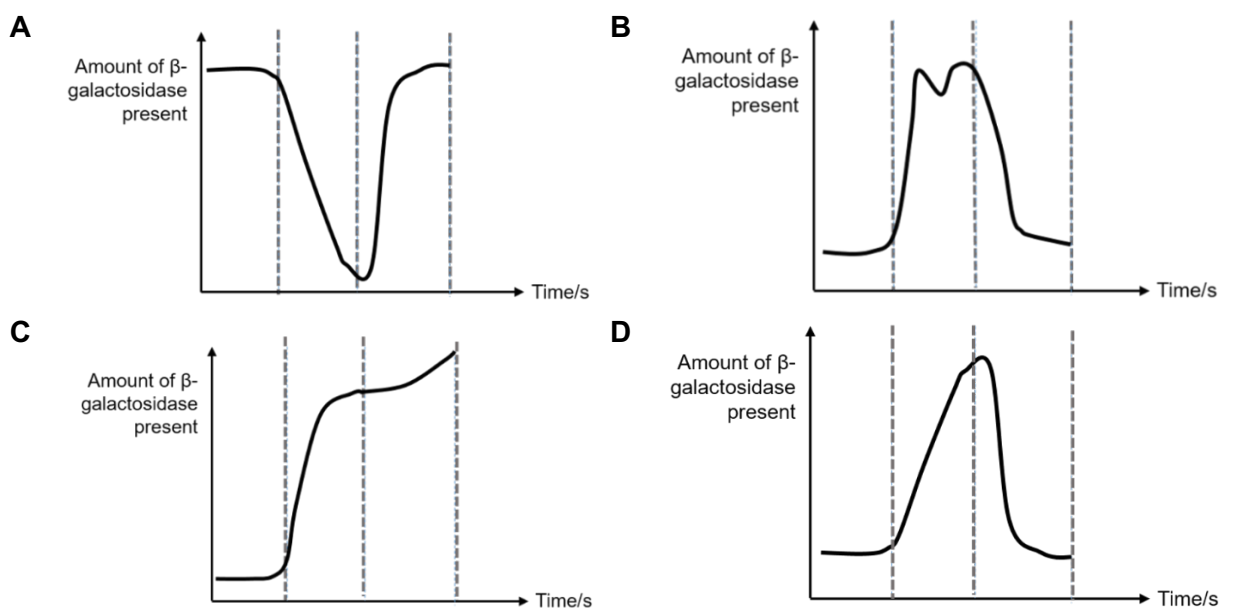
- 1 Conjugation
- 2 Transduction
- 3 Transformation

- A** 3 only
- B** 1 and 2
- C** 1 and 3
- D** 1, 2 and 3

- 15 *Escherichia coli* are able to metabolise both glucose and lactose for their energy requirement. In an experiment, researchers added glucose and lactose into the *E. coli* culture at different time points and measured the amount of β -galactosidase, glucose and lactose levels at regular time intervals. The arrows in the diagram indicate the addition of the respective metabolite into the culture.



Which graph correctly shows the corresponding amount of β -galactosidase present in the culture?



16 Which of the following statement(s) concerning *trp* operon is / are true?

- 1 A mutation of the regulatory gene leads to constitutive production of tryptophan.
- 2 There is one start and one stop codon in the mRNA of *trp* operon.
- 3 The repressor is inactive in the presence of excess tryptophan.
- 4 The mRNA codes for multiple polypeptides involved in the synthesis of tryptophan.

- A** 1 only
B 1 and 4
C 2 and 3
D 1, 2 and 4

17 A tall, green-stemmed plant with the genotype TTrr was crossed to a short, red-stemmed plant with the genotype ttRR. The F₁ plants were allowed to self-fertilise. A χ^2 test was carried out on the results obtained for the F₂ generation.

Part of the table of values for χ^2 is shown.

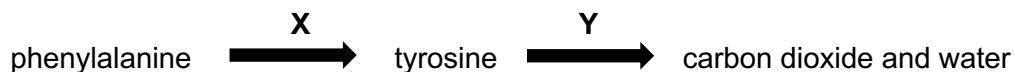
degree of freedom	probability				
	0.5	0.1	0.05	0.01	0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46

The value of χ^2 in this investigation was 7.6.

What is the probability of this value of χ^2 value?

- A** between 0.001 and 0.01
B between 0.01 and 0.05
C between 0.05 and 0.1
D less than 0.05

- 18** The following reaction sequence occurs in humans.



Genetic disease **P** is caused by an enzyme deficiency in step **X** and genetic disease **Q** is caused by an enzyme deficiency in step **Y**. Both conditions are rare and are caused by recessive alleles.

A person with genetic disease **P** marries a person with genetic disease **Q**.

Which phenotypes would be expected for their children?

- A** All have neither genetic disease
 - B** All have genetic disease **P** only
 - C** All have genetic disease **Q** only
 - D** All have both genetic diseases
- 19** A tall, pink-flowered plant is self-fertilised and produces the offspring shown.

	flower colour		
	red	pink	white
tall plants	73	157	67
dwarf plants	21	53	25

When self-fertilised, which type of plant will only produce identical offspring?

- A** dwarf, pink-flowered
- B** dwarf, red-flowered
- C** tall, red-flowered
- D** tall, white-flowered

- 20** Mammalian liver cells were homogenised, and the resulting homogenate centrifuged. Portions containing only mitochondria and cytosol (residual cytoplasm) were each isolated. Samples of each portion, and of the complete homogenate, were incubated in four ways:

- 1 with glucose
- 2 with pyruvate
- 3 with glucose plus cyanide
- 4 with pyruvate plus cyanide

Cyanide inhibits carriers in the electron transport chain. After incubation, the presence or absence of carbon dioxide and lactate in each sample was determined. The results are summarised in the table below.

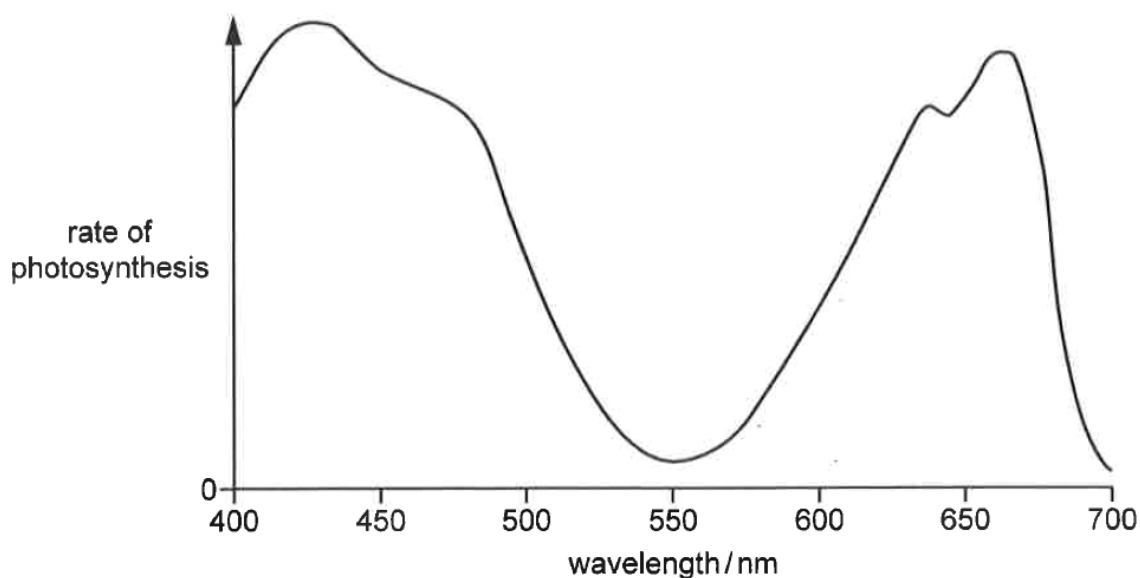
Tube	samples of homogenate					
	complete		mitochondria only		cytosol	
	carbon dioxide	lactate	carbon dioxide	lactate	carbon dioxide	lactate
1	✓	✓	✗	✗	✗	✓
2	✓	✓	✓	✗	✗	✓
3	✗	✓	✗	✗	✗	✓
4	✗	✓	✗	✗	✗	✓

✗ = absent, ✓ = present

Which statement can be concluded from the table?

- A** Both aerobic and anaerobic respiration were occurring in tube 3.
- B** Carbon dioxide was not formed when mitochondria were incubated with glucose as there was no oxygen present.
- C** Lactate formation in mitochondria was inhibited by the presence of cyanide.
- D** The action of cyanide would be similar to that of a cell experiencing anaerobic respiration.

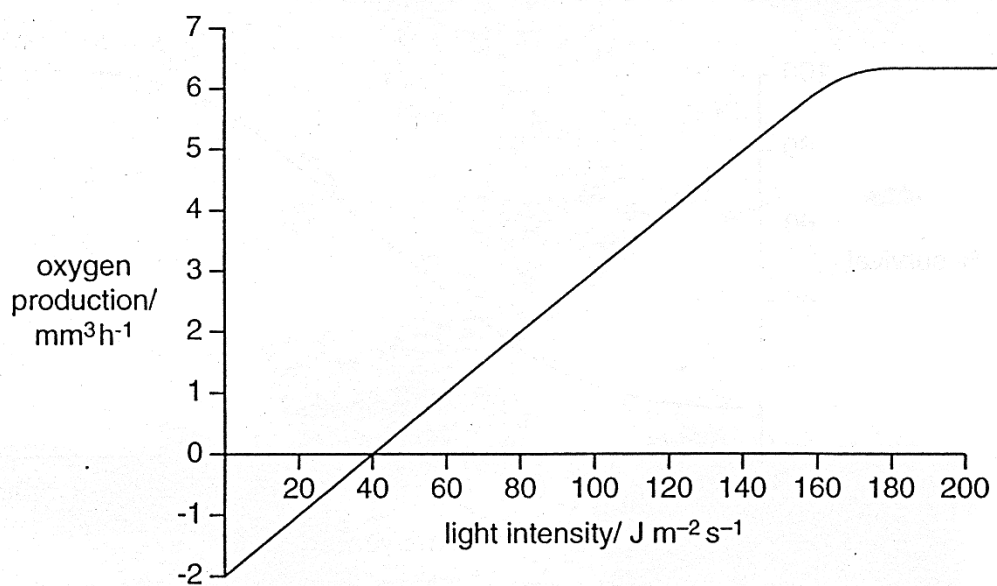
- 21** Which statement about respiration under anaerobic conditions in yeast and mammalian muscle tissue is correct?
- A** During respiration under anaerobic conditions, only a small proportion of the energy in the initial reactants is used for ATP production.
 - B** Under anaerobic conditions, ethanol production releases fewer carbon dioxide molecules than lactate production for the same yield of ATP.
 - C** The amount of energy yielded from lactate production and ethanol production cannot be compared since the initial reactants are different.
 - D** The continued production of ATP depends on respiration of ethanol in yeast and respiration of lactate in mammalian tissue.
- 22** The graph below shows the action spectrum for photosynthesis of a particular plant.



Which of the following explains the rate of photosynthesis for this plant at wavelengths of light between 525 nm and 575 nm?

- A** Chlorophyll a is unable to undergo photoactivation between this wavelength.
- B** Energy level of light between this wavelength is low.
- C** The plant does not have pigments that absorb green light.
- D** The plant lacks carotenoid and xanthophyll pigments.

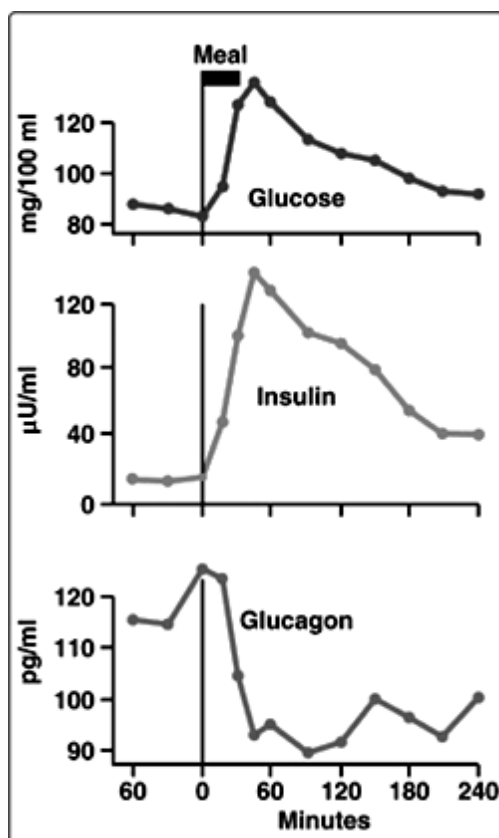
- 23** The graph shows the relationship between oxygen production in photosynthesis and light intensity for a unicellular green organism in 0.02% sodium hydrogencarbonate solution.



The most likely explanation of the fact that the graph levels off at $180 \text{ J m}^{-2} \text{s}^{-1}$ is that the system is

- A** light limited and carbon dioxide saturated.
- B** light limited and the temperature is below optimum.
- C** light saturated and carbon dioxide limited.
- D** light saturated and the temperature is above optimum.

- 24 The figure shows the levels of glucose, insulin and glucagon found in blood, before and after a carbohydrate-containing meal was ingested.



Which row correctly identifies the events occurring at the respective timings?

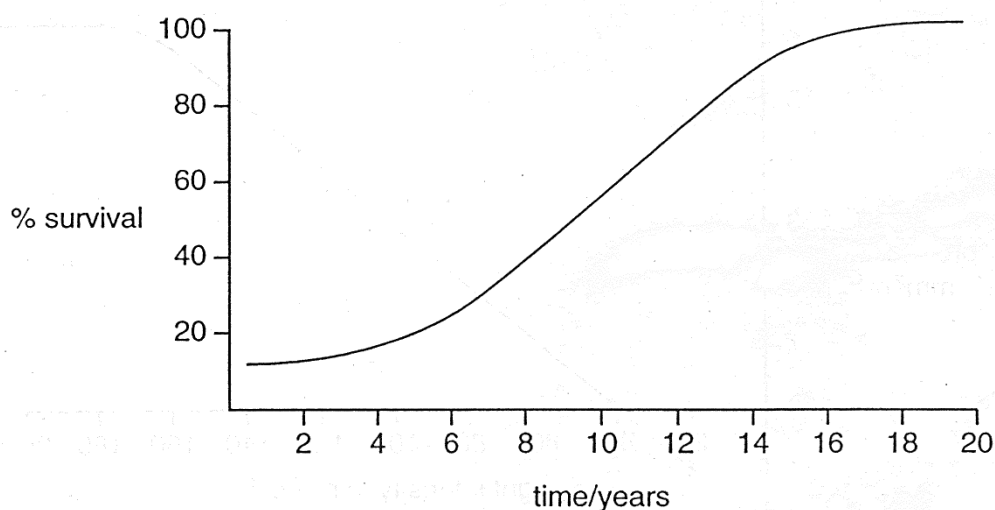
	Timing	Events
A	60 minutes before meal	Glucagon results in the activation of glycogen phosphorylase, leading to increased rates of glycogenolysis in liver cells and an increase in blood glucose levels.
B	30 minutes before meal	Glucagon binds to G protein-linked receptors, leading to the activation of G proteins due to the phosphorylation of GDP molecules. Cellular responses lead to the maintenance of blood glucose levels.
C	At the start of meal (0 minutes)	Insulin binds to tyrosine kinase receptors, resulting in the cross-phosphorylation of tyrosine residues. Cellular responses lead to a decrease in blood glucose levels.
D	60 minutes after meal	Insulin results in the translocation of glucose transporters to the cell surface, leading to increased rates of glucose uptake in muscle cells and a decrease in blood glucose levels.

25 Which statements are correct interpretations of Darwinian evolutionary theory?

- 1 Advantageous behaviour acquired during the lifetime of an individual is likely to be inherited.
- 2 In competition for survival, the more aggressive animals are more likely to survive.
- 3 Species living in a stable environment will not evolve any further.
- 4 Variation between individuals of a species is essential for evolutionary change.

- A** 1, 2 and 4 only
B 2 and 3 only
C 3 and 4 only
D 4 only

26 The graph shows the effect of pesticide treatment on houseflies over a number of years. A standard amount of pesticide was used each year in summer.



How is the effect of the pesticide best explained?

- A** A few resistant flies reproduced more successfully, and the resistance allele increased in frequency.
B At every generation an increasing proportion of flies mutated to become resistant.
C Repeated exposure to the pesticide caused the flies to become more resistant.
D The allele for resistance mutated from the recessive form to the dominant form.

- 27** Reproduction in seahorses, *Hippocampus*, is unusual as it is the male rather than the female that becomes pregnant. The male has a brood pouch located on its tail. The larger the male the larger the pouch. The female transfers unfertilised eggs into the pouch. The larger the female the more eggs are produced that can be transferred to the brood pouch. The male releases sperm onto the eggs and they are fertilised. The male carries the developing brood for a period of several weeks until he finally gives birth.

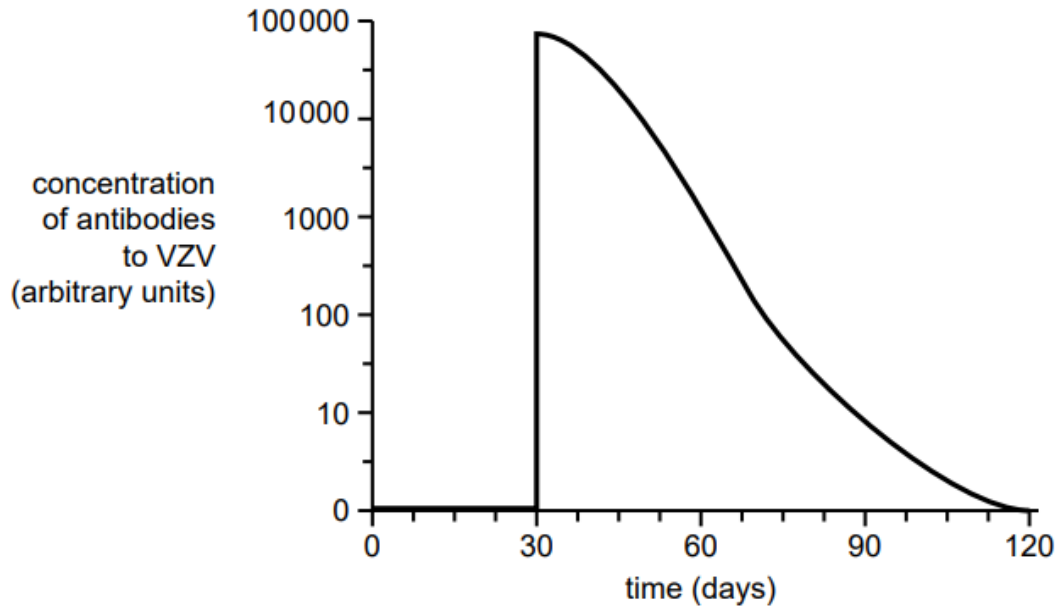
Research into seahorse populations has revealed the following.

- They are monogamous. A male and female remain together for the whole mating season.
- Within a population, mates are selected by size. Large females mate with large males and small females mate with small males.
- Few intermediate sized individuals are produced and they have a low survival rate.

Which of the following conclusions **cannot** be drawn from the above information?

- A** Disruptive selection has occurred.
- B** Intermediate sized individuals are selected against.
- C** There is phenotypic variation in tail size in males.
- D** Monogamy results in lower selection pressure as males and females remain together the whole mating season.
- 28** An antibody is a protein complex secreted by plasma B cells. Which of the following statements regarding the mechanisms of generating antibody diversity is true?
- A** An antibody changes its affinity upon binding to the antigen through a process known as somatic hypermutation.
- B** Class switching occurs to produce antibodies with different constant regions in B cells undergoing differentiation in the bone marrow.
- C** DNA rearrangement during somatic recombination generates a wide repertoire of B cells with different B cell receptors.
- D** During clonal expansion, rearrangement of the V, D and J segments give rise to antibodies with different variable regions.

- 29** Chickenpox (varicella) is a highly contagious viral disease caused by the varicella-zoster virus (VZV). A laboratory technician measured the concentration of antibodies to VZV in a person's blood over a 120-day period. An event occurred on day 30 that significantly altered the concentration of antibodies. The concentration of antibodies over the 120 days is displayed in the graph below.



Which one of the following events could have occurred on day 30?

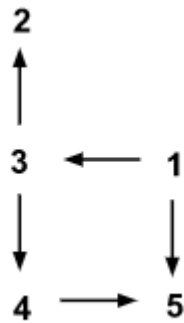
- A** a booster vaccination against VZV for the person
- B** an exposure of the person to VZV
- C** an injection of antibodies to VZV into the person
- D** an oral dose of antibiotics was given to the person

- 30** Investigations into the possible current and future impacts of climate change need to be put into the context of the well-documented and on-going impacts of other drivers of change, such as population growth. The statements below are effects of climate change and population growth.

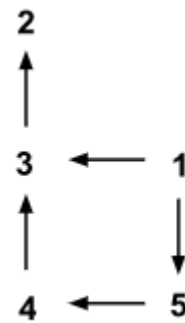
- 1 More intensive land use results in degradation of soils and more rainfall run-off
- 2 Food and feed shortages
- 3 Greater frequency of water deficit in soil for crops and pasture growth
- 4 More frequent dry years experienced
- 5 Increase food demand and competition for pastures

Which of the following diagram correctly illustrates the relationship between climate change and population growth?

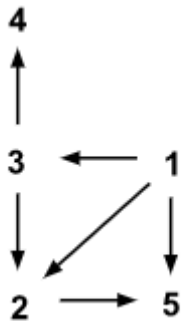
A



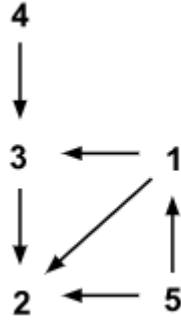
B



C



D



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